Pencil upper gate: interim checkpoint 9 observations

Run: pencil-p186379-20261030-c1. Caller-supplied source commit: 543a54b4b1c3be7052872af3ffe393a44c69a3c3.
Observations only. No statistical, speed, mathematical or timing qualification is inferred.
One public synthetic mathematical target at p=186379. Interim snapshot of a running computation; no complete measured comparison.
Workloads: 1; expected pairs per price cell: 5; slots: 18; observed: 4; missing: 14.

Recorded outcomes

pencil:timeout: 1
rho:verified: 2
rho_AA:verified: 1
Missing rows remain missing; capped, timeout, error and other outcomes retain their original labels. Counts from different epochs are not pooled.

Pair coverage by price cell

Workloads: 1; expected pairs per price cell: 5; slots: 18; observed: 4; missing: 14.

Arms: pencil, rho, rho_AA. Planned slots are not observations.

Fewer than two comparable ratio cells are present. Diagram of planned scope and observed progress; no performance plot.
Observations only. This is an interim checkpoint report, not terminal gate accounting, a mathematical verdict, or an algorithm ranking.
The versioned generation9 archive, manifest and selected session files passed hash/size checks. The embedded session remains running with a null final records digest.
Four of18 planned execution slots are retained: two control warmups report verified; pencil warmup reports timeout with watchdog expired and signal9; one rho measured round reports verified. Producer statuses are not independent scientific replay.
Fourteen planned slots remain absent in this checkpoint;14of15 measured slots are absent. No measured pencil row and no complete candidate/reference pair exist. All36 cost-ratio cells are null; absent work is not a zero-cost outcome.
All clocks remain unqualified. No fresh benchmark epoch or independent scientific replay is established by this packet.
All five required measured pairs remain unavailable.
Original scientific protocol version4 and unchanged source/binary/inputs govern data. Version5 only amends future same-request controller supervision. The six-hour budget,08:10:24 UTC scientific cutoff and11:17:54 UTC outer expiry are unchanged.
Controller and worker resource observations remain separate from finite algorithm counts. Execution is ongoing, so total active time and total spending remain unknown; prior failed attempts and costs remain separately retained.
Native canonical packaging and checking retain expected incompleteness findings. The final post-restoration check has five shortfall findings and no resource-binding error.
A resource metadata update raced packaging. Original exact bytes were restored; newer support is separately retained.
Neither raw scientific rows nor the frozen analysis changed.

Separate resource observations

No conversion to algorithm cost units.

{"accounting_notes":["Worker time continues across controller ownership changes; controller invocations are separately retained.","G9 includes a producer-recorded pencil warmup timeout; its raw rusage/status remain in the archived record, not converted to a complete algorithm cost.","No complete measured pair or independently replayed candidate exists in this snapshot.","No total active time or total spend is claimed while execution and terminal accounting remain incomplete.","Prior budgets and failed attempts remain separate, without refunds or pooled result rows."],"allocated_dollars":null,"allocation_and_algorithm_costs_distinct":true,"charged_categories":["candidate","rho","rho_AA","warmups","computation","fresh_epoch_if_any","independent_replay_if_any","recovery","failed_attempts"],"controller_amendment":{"path":"/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/protocol_draft/replacement_upper_gate_20261030/protocol/controller-continuation-v5.json","sha256":"d00be8a9e4b12fd4e6bb60d3d6c423b9213850de1859e22e248b52cddf70c956"},"controller_build":"isolab-recovery-controller:592e9381-e505-48bc-a796-3d02c4d3a922","controller_continuation_admission":{"path":"/home/ec2-user/cr

ypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/protocol_draft/replacement_upper_gate_2026
1030/controller-takeover/continuation-admission.json","sha256":"187f21544280fdef4835b52e1ea860a1ed8472e0f3b50b2
a6b6fdfd5da215ca2"},"controller_end_utc":null,"controller_history":{"path":"/home/ec2-user/crypto-pencil-signed-
graph-20261029/research/pencil_large_prime_20261029/protocol_draft/replacement_upper_gate_20261030/controller-tak
eover/lineage-observation.json","sha256":"79d3b325cea11f33135ff741846e02a725e4e520a9909de5f192461db53c24a6"},"c
ontroller_invocations_observed":2,"controller_phase":"BUILD","controller_status":"IN_PROGRESS","cost_conversion":
null,"cumulative_active_worker_seconds_limit":21600,"failed_costs_retained":true,"maximum_active_scientific_payloa
ds":1,"missing_costs_not_zero":true,"observation_utc":"2026-10-30T06:51:07Z","original_authorization_start_unix":
1793337474,"outer_D_unix":1793359074,"prior_b1_resources":{"path":"/home/ec2-user/crypto-pencil-signed-graph-2026
1029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-b1/terminal-partial/native-analysis/resourc
e-observations-worker-terminal.json","sha256":"4880cc449b6d4b0065cd71d1be119cb748bb5f658d8574b87051454738b6f928
"},"request_G_unix":1793347824,"resource_authorization":{"path":"/home/ec2-user/crypto-pencil-signed-graph-20261
029/research/pencil_large_prime_20261029/protocol_draft/replacement_upper_gate_20261030/resource_authorization.jso
n","sha256":"5ea8e080af4b28572f3ff1d074e9e261ef8ad3c24a53a6e7d06d1b37b0316d25"},"schema":"pencil-interim-resour
ce-observations/v1","scientific_submission_count":1,"scientific_total_cost":null,"scope":"Interim checkpoint9 dat
a and separately timestamped live controller observation. Run remains active; this is not terminal allocation
or total scientific cost accounting.","worker_active_seconds":null,"worker_attempt_observed":0,"worker_elapsed_se
conds":null,"worker_instance_id":"i-04d58c2b309fb6549","worker_instance_terminated_verified":false,"worker_pod_ui
d":"3a7ab215-49d1-4982-8c78-5a0c71646842","worker_restores_observed":0,"worker_termination_verified":false}

All36 exact price cells

S2 is twice S/M; N10 is ten times the native event price. Incomplete/unavailable ratios are null, not zero. Partial admitted sums are retained in JSON as diagnostic only; no partial sum is promoted to a complete ratio.

Cell	S2/I/N10	Admitted/expected	Method/rho exact	With calibration/rho exact
1	0/1/0	0/5	null	null
2	0/1/1	0/5	null	null
3	0/1/10	0/5	null	null
4	0/10/0	0/5	null	null
5	0/10/1	0/5	null	null
6	0/10/10	0/5	null	null
7	0/100/0	0/5	null	null
8	0/100/1	0/5	null	null
9	0/100/10	0/5	null	null
10	0/1000/0	0/5	null	null
11	0/1000/1	0/5	null	null
12	0/1000/10	0/5	null	null
13	1/1/0	0/5	null	null
14	1/1/1	0/5	null	null
15	1/1/10	0/5	null	null
16	1/10/0	0/5	null	null
17	1/10/1	0/5	null	null
18	1/10/10	0/5	null	null
19	1/100/0	0/5	null	null
20	1/100/1	0/5	null	null
21	1/100/10	0/5	null	null
22	1/1000/0	0/5	null	null
23	1/1000/1	0/5	null	null
24	1/1000/10	0/5	null	null
25	2/1/0	0/5	null	null
26	2/1/1	0/5	null	null
27	2/1/10	0/5	null	null
28	2/10/0	0/5	null	null
29	2/10/1	0/5	null	null
30	2/10/10	0/5	null	null
31	2/100/0	0/5	null	null
32	2/100/1	0/5	null	null
33	2/100/10	0/5	null	null
34	2/1000/0	0/5	null	null
35	2/1000/1	0/5	null	null
36	2/1000/10	0/5	null	null

Evidence bindings

/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/checkpoint-g9-observations/native-analysis/native-analysis.json
4f439949a2040d6b88cf9969c76316dfe7384f4963e75efba58c91934c83cd69
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/checkpoint-g9-observations/independent-observation-review.json
8c85f286811691d53322fd2131b69ab6eab1711c0a41eed9f3501663dc41169d
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/checkpoint-g9-observations/g9-local-custody.json
93c82f0f7a39781e2ce9a226f46a3fbf58a49be4555b26e44e32af13bb264d62
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/checkpoint-g9-observations/g9-observed-state.json
cfa9916b910dbe3fe42105437a6fec77852f2591ed7fcbcae2d1988424950788

/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/checkpoint-g9-observations/native-analysis/execution-history.json
8aabb530ebee7515f41917e4606f90accb289d758ba9a19b01aa5440d6c2dc4b
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/checkpoint-g9-observations/native-analysis/export.json
7d7955749d0e8be7f93f68d6b20fe141cef8db53784f21482dadc6d096c72791
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/checkpoint-g9-observations/native-analysis/resource-observations-supported-update.json
0cea563aa6dea9791792534f725c31369df13128bcbae90f17fb64cc0afdc9f8
/home/ec2-user/crypto-pencil-signed-graph-20261029/research/pencil_large_prime_20261029/runs/p186379-gate-20261030-c1/checkpoint-g9-observations/native-analysis/resource-observations.json
0f25401a77a5e09e3f2da704e3189f1a93920fbf47a3c70075f6834ee07de181